

Satellite Image Analysis-based Flood Prediction and Monitoring System using Deep ConvNeural Network

by

Sajibul Islam (213002134)
Tamimur Rahman (213002224)
Md: Osman Gani (213002235)

*Project (CSE 400) submitted in partial fulfillment of the
requirements for the degree of*

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Under the supervision of

Dr. Muhammad Aminur Rahaman



DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
GREEN UNIVERSITY OF BANGLADESH

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DECLARATION

Title Satellite Image Analysis-based Flood Prediction and Monitoring System using Deep ConvNeural Network
Authors *Sajibul Islam, Tamimur Rahman, and Md: Osman Gani*
Student IDs 213002134, 213002224, and 213002235
Supervisor Dr. Muhammad Aminur Rahaman

We declare that this project entitled *Satellite Image Analysis-based Flood Prediction and Monitoring System using Deep ConvNeural Network* is the result of our own work except as cited in the references. The project has not been accepted for any degree and is not concurrently submitted in candidature of any other degree.

Sajibul Islam
213002134

Tamimur Rahman
213002224

Md: Osman Gani
213002235

Department of Computer Science and Engineering
Green University of Bangladesh

Date: July 18, 2025

CERTIFICATE OF ENDORSEMENT

This is to certify that the project entitled *Satellite Image Analysis-based Flood Prediction and Monitoring System using Deep ConvNeural Network* by **Sajibul Islam** (Student ID: 213002134), **Tamimur Rahman** (Student ID: 213002224), and **Md: Osman Gani** (Student ID: 213002235) has been successfully defended and examined. Based on the evaluation and discussion held on **July 18, 2025**, we, the undersigned, hereby endorse the project for acceptance and approval.

Md. Al Imran
External Member, Board: 07

Md. Solaiman Mia
Board Member, Board: 07

Assistant Professor
Department of Computer Science and Engineering
Green University of Bangladesh

Mr. Syed Ahsanul Kabir
Chair, Board: 07

Associate Chairperson of CSE & Associate Professor
Department of Computer Science and Engineering
Green University of Bangladesh

Date: July 18, 2025

CERTIFICATE

This is to certify that the project entitled **Satellite Image Analysis-based Flood Prediction and Monitoring System using Deep ConvNeural Network**, submitted by **Sajibul Islam** (Student ID: 213002134), **Tamimur Rahman** (Student ID: 213002224), and **Md: Osman Gani** (Student ID: 213002235), undergraduate students of the **Department of Computer Science and Engineering**, has been examined. Upon recommendation by the examination committee, we hereby accord our approval of it as the presented work and submitted report fulfill the requirements for its acceptance in partial fulfillment for the degree of *Bachelor of Science in Computer Science and Engineering*.

Dr. Muhammad Aminur Rahaman

Department of Computer Science and Engineering
Green University of Bangladesh

Dr. Md Ahsan Habib
Chairperson

Department of Computer Science and Engineering
Green University of Bangladesh

Place: Narayanganj

Date: July 18, 2025

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Sajibul Islam, Tamimur Rahman and Md: Osman Gani

Green University of Bangladesh

Date: July 18, 2025



Dedicated to
my loving Mother

– *Sajibul Islam*

To my beloved parents
the source of my inspiration

– *Tamimur Rahman*

For the man who taught
me to dream,
This is for you, Abbu
Your absence is felt, your
love remains.

– *Md: Osman Gani*

ABSTRACT

Floods are globally one of the most devastating natural disasters , timely flood detection is a critical component of effective disaster management and response systems. Traditional flood assessment techniques often rely on manual interpretation of satellite data, which is time-consuming, labor-intensive, and subject to human error. To address these challenges, this research proposes a deep learning-based framework for automated flood prediction using satellite imagery. The model is built using a customized Convolutional Neural Network (CNN) architecture that integrates multiple convolutional layers, Batch Normalization, pooling layers, and GlobalAveragePooling, followed by dense and dropout layers for robust classification. The model employs image preprocessing techniques such as band stacking, normalization, and resizing, along with data augmentation strategies to improve generalization and handle class imbalance. To ensure reliable evaluation, the dataset is split into training, validation, and testing sets, with a focus on the binary classification of flooded vs. non-flooded regions, the proposed CNN model achieves an overall accuracy of 80.6% and demonstrates strong performance in terms of precision (98.9%) and F1-score (85.9%) on the validation set. Analysis is supported through metrics such as AUC, loss curves, and confusion matrices. The results confirm that the CNN-based approach effectively learns spatial patterns from satellite data, providing a scalable and accurate method for flood area identification. With further enhancements, this system holds significant potential for integration into real-time monitoring platforms, offering valuable support in early warning system before floods, emergency response, and climate resilience efforts.

Keywords: Flood , Deep Learning , DCN Flood Prediction , Satellite image

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1. Introduction

1.1 Problem Statement

Flood forecasting, while vital, still suffers from a series of necessary limitations that hinder the creation of efficient and timely forecasting systems [1]. Classical flood monitoring often depends on separate sources of information like rain gauges, river stages, or minimal meteorological inputs which are inadequate representations of the complex and dynamic nature of flood phenomena. Satellite remote sensing has been seen as a potential leader in overcoming this gap. Yet even so, the integration of high-resolution multi spectral images by machine learning models is a gigantic problem. Satellite data can be different from each other over time, resolution, and sensor type. Aligning them and extracting effective flood features from this heterogeneous input is no small task [2]. Furthermore, flood patterns are hard to label making proper classification tricky.

This project aims to outsmart these challenges by:

- Creating a deep learning-based system that uses multispectral satellite images to detect and predict flood-prone zones using spatial feature analysis [3].
- Using multi spectral satellite imagery (worldview3, Maxarbd) It captures surface features water bodies, crops, roads, and more before and after floods. A full picture, every time [4].
- Visualizing predictions clearly. Maps, zones, overlays whatever makes it easier to understand. Because people on the ground need to trust what they're seeing [5].

Figure 1.1 presents a side-by-side comparison of satellite images captured before and after a flood. In the images on the left, the terrain looks dry, with well-defined river outlines. However, in the right-hand images taken after the flood, there is a noticeable spread of blue-toned areas, showing how floodwaters have spilled over into nearby lands. These dark blue zones highlight areas submerged under water, indicating the extent of the flooding.

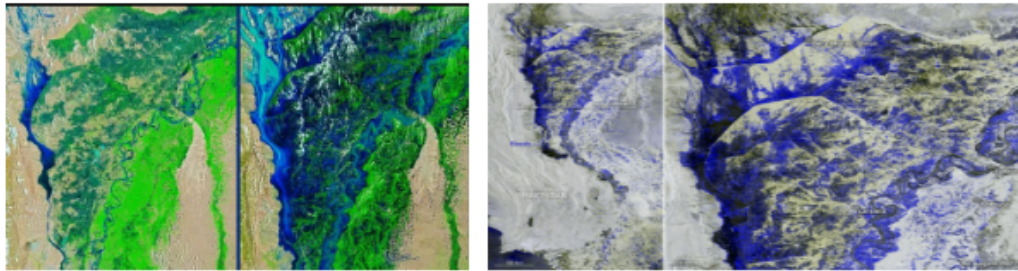


Figure 1.1: Satellite imagery before and after flooding

1.2 Motivation

The motivation behind this project is:

- **Humanitarian Need:** Floods displaced over 12.6 million people in 2023 (IDMC). This phenomenon tells us to make a solution that would be fast and early warning [6].
- **Technological Imperative:** Manual analysis of satellite imagery takes too much time, is subjective, and can delay decision-making during disasters [7].
- **Data Availability:** Open-access sources like Maxar's xBD dataset (0.5m/pixel imagery) are suitable for training AI models since they provide high-resolution images [8].
- **Climate Resilience:** Flooding is becoming more common because of climate change. Nowadays, automated AI solutions are emergency needs for improving early forecasts of flooding, which will help us to deal with floods [9].

1.3 Overview

Floods are among the most destructive natural disasters. They cause significant damage to infrastructure, smash economies, and force peoples to leave places. Traditional flood monitoring systems depend on site sensors and manual image analysis. These methods are often expensive, limited in coverage, and slow to respond during critical moments. With the growth of remote sensing technologies and artificial intelligence, automated flood detection has become more practical and scalable. This project uses high-resolution satellite imagery from Maxar's xBD dataset along with various topographic data, such as digital elevation models (DEMs) and hydrological features, to create an accurate, data-driven flood prediction system. Central to this work is a deep learning model suggested particularly for flood detection. We suggest a low-weight and computationally cheap convolutional neural network (CNN) architecture optimized for Maxar imagery. Our custom architecture, specially designed, differs from conventional pre-trained models and helps reduce overfitting as well as improve computational efficiency. Furthermore, the model also integrates multi-scale feature extraction mechanisms such as dilated convolutions to capture fine-grained hydrological patterns of flooding (e.g., water accumulation around structures) and large-scale hydrological patterns across regions. Employing synergistic computer vision and geospatial processing, our model enhances rapid and precise flood prediction, which could significantly enhance early warning systems and disaster response.

1.4 Project Questions

This project addresses the following key questions:

1. Can we obtain satellite images for flood prediction?
2. Can preprocessing of satellite images be important for flood prediction?
3. Can we remove unwanted elements like clouds from satellite images?
4. Can satellite images be effectively used and analyzed for flood prediction?

1.5 Objectives

- To create an application that can easily predict floods from satellite images.
- To develop and implement preprocessing techniques to enhance the quality and usability of satellite images for flood prediction.
- To design, develop, and evaluate deep learning models capable of accurately analyzing preprocessed satellite images for effective flood prediction.
- To ensure the developed flood prediction model is robust, generalizable across diverse environmental conditions, and provides interpretable insights into its predictions.

1.6 Contribution

- An efficient API will be developed to enable seamless integration between different components of the system.
- A powerful deep learning model will be created to predict floods accurately using satellite image analysis.
- Satellite images will be integrated into the system to support flood prediction.
- The proposed system is expected to outperform existing models.

1.7 Report Structure

This report is structured as follows:

- **Chapter 1: Introduction** – Introduces the flood detection research topic, explains the motivation behind the study, outlines the main objectives, and highlights the contributions of this work. It sets the foundation for understanding the importance of using satellite imagery and deep learning for flood prediction.
- **Chapter 2: Literature Review** – Reviews key studies on flood detection using satellite imagery and deep learning techniques, including CNN architectures, multispectral and SAR data, and the challenges faced in this domain.

- **Chapter 3: Web Application and System Integration** – Details the development of the web-based interface and API service and user interaction with the system.
- **Chapter 4: Methodology** – Describes the design and development of the flood prediction system, including dataset preparation, model architecture, training pipeline, and evaluation metrics.
- **Chapter 5: Results and Discussion** – Presents experimental results, performance evaluations, confusion matrix analysis, and comparisons of different deep learning models used for flood detection.

2. Literature Review

2.1 Introduction

This part of the literature review presents key studies on flood detection using satellite imagery and deep learning methods. Review highlights techniques making use of multispectral imagery, particularly that of Sentinel satellites, in identifying flooded and non-flooded regions [10]. Convolutional Neural Networks (CNNs) have been widely used since they perform optimally in the extraction of spatial patterns from remote sensing data [11]. Several preprocessing techniques, spectral band ratios, and classification models were experimented and assessed for their accuracy and applicability. Current studies also highlight the importance of automated and scalable tools for flood monitoring in data-scarce or fast-changing situations.

2.2 Pertinent Studies on Flood Detection with Deep Learning

Deep learning has recently brought big improvements in flood detection, especially when used with satellite images. Many studies have shown that deep neural networks can accurately find flooded areas from space.

One study proposed a Vision Transformer (ViT) model that works with both Sentinel-1 (SAR) and Sentinel-2 (optical) satellite images from the SEN12-FLOOD dataset [12]. The ViT model gave strong results 84.84% accuracy with Sentinel-1 and 83.14% with Sentinel-2. It worked better than regular CNN models [13].

Another study introduced PSO-UNET, which combines Particle Swarm Optimization (PSO) with the popular UNET model for detecting flash floods [14]. Using 984 Sentinel-2 images from Vietnam, the model achieved a Dice Score of

over 79.75%, which was 8.59% better than the normal UNET. This improved the detection of flooded areas.

In another research, Sentinel-1 SAR images were used for mapping urban floods. Extra data like road closures and rainfall reports were also used [15]. Several machine learning methods were tested. The Support Vector Machine (SVM) and an unsupervised Change Detection (CD) method worked best, giving up to 87% accuracy. Random Forest (RF) and Maximum Likelihood Classifier (MLC) also gave good results (0.69 and 0.83 accuracy). Among them, the CD model performed very well.

Another study focused on using both VV and VH SAR images from Sentinel-1A with a Multi-Scale Deeplab (MS-Deeplab) model [16]. This model used MobileNetV2 and DeeplabV3+ to handle different image scales and polarizations. It did better than the original DeeplabV3+ model, with a 3.27% improvement in IoU and 1.69% higher Pixel Accuracy.

Finally, one study built a near-real-time flood detection system using Sentinel-1 SAR images and elevation data (ALOS DEM) from the Yangtze River area [17]. Their CNN-based model worked faster and more accurately than older threshold-based methods. VH polarization gave the best results, and the DEM data added only a small improvement.

2.3 Convolutional Neural Network (CNN) Based Image Classification

Convolutional Neural Networks (CNNs) are a popular type of deep learning model made for working with image data. These models automatically learn patterns and important features in images by using small filters, called kernels. CNNs are used in many fields like medical imaging, face detection, remote sensing, and even natural disaster monitoring because they can understand complex image details [18]. When it comes to satellite images, CNNs work really well. They can pick up small and large patterns from the image, even if the important features are hard to see or spread across many color bands. This makes CNNs useful for identifying land types, water bodies, and flooded areas.

2.3.1 Applications of CNNs for Flood Detection

CNNs have shown great promise in mapping floods using satellite data. Many research projects have used CNN models to tell the difference between flooded and non-flooded areas.

One study used a Multi-Scale Deeplab CNN model on Sentinel-1 SAR images with both VV and VH polarizations. The model learned useful features from the images and performed well across different locations and flood conditions [19].

Another study showed how CNNs can be used for fast, real-time flood detection in the Yangtze River Basin using Sentinel-1 SAR images [20]. The CNN model gave better results than older threshold-based methods and was also faster, which makes it more useful for emergency response systems.

2.4 Challenges in Flood Detection Using CNNs

Even though CNN-based models are powerful tools for detecting floods from satellite images, they still face some important challenges. These problems can affect how well the models work in real-life situations.

2.4.1 Heterogeneity of the Dataset

Flood detection datasets are usually made up of images taken under many different conditions. These differences include changes in weather, seasons, locations, and types of satellite sensors (like optical vs. SAR) [21]. Because of this, the image quality, water visibility, cloud cover, and vegetation can vary a lot from one image to another. This variety makes it hard for deep learning models, like CNNs and Vision Transformers, to learn patterns that work well everywhere [22]. For example, a model trained only on optical images may not perform well if clouds are blocking the view. On the other hand, SAR images can still work through clouds but come with their own problems, like noise from the terrain. To handle this, models need to be trained on a wide range of data so they can work reliably in different situations.

2.4.2 Environmental Conditions and Noise

Satellite images can be affected by environmental factors like cloud shadows, fog, sensor noise, and light distortion. These issues can hide the flooded areas or make

dry areas look flooded. This often causes the model to make wrong predictions. These problems are worse with optical images because they rely heavily on clear weather and light. To reduce the effect of such conditions, researchers like [23] trained a Vision Transformer using both Sentinel-2 optical and Sentinel-1 SAR images. Since SAR images are less affected by clouds, this helped make the model more reliable under different environmental conditions.

2.5 State-of-the-Art Techniques in Flood Detection

Recent advances in remote sensing and deep learning have led to the emergence of various state-of-the-art techniques for flood detection using satellite imagery. These methods aim to enhance classification accuracy, address environmental heterogeneity, and reduce training time. Key methodologies explored in contemporary research are discussed below.

2.5.1 Hybrid CNN and Traditional Machine Learning

Some research studies have contrasted and evaluated the traditional image processing techniques with deep learning techniques for flood detection for example, contrasted the global threshold technique with a variety of famous deep learning models based on their performance in near-real-time flood detection. The results showed that while the global threshold technique is efficient and simple to use in large-scale and near-real-time flood detection, it is not competent in complex nonlinear classification tasks and is sensitive to noise. On the other hand, the deep learning architectures experimented with by [24], FCN-8, SegNet, UNet, and DeepResUNet, were more accurate and reliable in segmenting flooded regions. These models are better equipped to handle complex terrain and varied image conditions since they can learn contextual features and spatial patterns automatically from data. The study highlighted that deep learning approaches are particularly capable of bridging the limitations of threshold-based approaches. DeepResUNet emerged with a high performance-computational complexity trade-off from the models tested, attributed to its residual network and lower parameters.

2.5.2 Advanced CNN Architectures

A few of the recent deep models such as U-Net, ResNet, and DenseNet have transformed flood mapping into a much sophisticated area. U-Net, for instance, has become popular in pixel-wise segmentation and has been applied successfully in demarcating the flooded region from SAR images. For instance, [25] utilized U-Net, as well as other deep learning structures like DeepResUNet, to classify flooded regions from Sentinel-1 SAR images with great accuracy even in situations where SAR's benefits can be exploited, e.g., under cloud cover. Other architectures, generally with residual connections like ResNet50 and DenseNet121, have been used for their ability to promote feature propagation, depth efficiency, and robustness, proving to be effective especially on unbalanced flood datasets.

2.5.3 Data Augmentation Techniques

For improving model generalization and robustness, data augmentation methods are heavily applied in flood detection studies. The most common augmentation techniques include image rotation, flipping, brightness adjustments, and random cropping. These techniques simulate various environmental, seasonal, and lighting scenarios and allow the model to be trained on varied inputs. Such strategies, coupled with efforts to acquire large and far-reaching datasets from different sources and events, are crucial to overcome the limitations of limited training samples and environmental noise, as evident in research leveraging multi-modal data or combining large event-based data collections [26].

2.6 Gaps in the Current Literature

Even though many good studies have been done on flood detection using satellite images and deep learning, some important issues still remain. This research tries to solve the following key problems:

- **Small and Unbalanced Datasets:** Most existing models are trained on small or region-specific datasets. These datasets don't cover different types of flood situations, land types, or locations. As a result, models may not work well in new areas. In our work, we use a more balanced and diverse dataset that

combines images from both Sentinel-1 and Sentinel-2 satellites. This helps the model perform better in different conditions.

- **Low-Quality Labels:** Many earlier studies only use rough labels or large image tiles to identify flooded areas. This is not good enough for accurate flood detection. In this study, we use a technique called semantic segmentation (like U-Net) to create detailed, pixel-level flood maps that are much more accurate.
- **Limited Use of Mixed Satellite Data:** Some models are built for either optical images (like Sentinel-2) or radar images (like Sentinel-1), but not both. Very few studies combine these two types. In this research, we fuse data from both sensors to make flood detection more reliable, even in cloudy weather.
- **Lack of Explainability:** Most CNN-based models are like black boxes — they give predictions, but we can't see how or why. This makes it hard to trust the results. Our work includes tools like Grad-CAM and SHAP to highlight which parts of the image affect the model's decision, making it easier to understand.
- **Not Ready for Real-World Use:** Many models work well in lab tests but are too big or slow for use in real situations. This research focuses on building a smaller and faster model that can be used in real-time disaster warning systems and emergency response tools.

2.7 Conclusion of Literature Review

This literature review shows that deep learning, especially Convolutional Neural Networks (CNNs), has become a powerful tool for detecting floods using satellite images. Many studies have shown good results by using both radar and optical satellite data to identify flooded areas. Advanced models like U-Net, ResNet, and hybrid approaches (like CNN with SVM) have improved how accurately we can detect and separate flooded regions. However, there are still some common problems. Most datasets are small or unbalanced, with more non-flood images than flood ones. This makes it hard for models to learn well. Also, many models are trained for specific regions or seasons, so they may not work well in other places. Another big issue is that deep learning models often work like black boxes, we

don't always know why they make a certain prediction. Because of these challenges, there is a strong need for better flood detection systems that are more accurate, explainable, and able to work in different conditions. This research is focused on solving these problems by using better data, combining different satellite sources, and building models that are easier to understand and more useful in real-world situations.

2.8 Summary of Literature Review

Table 2.1 highlights the existing works, key contributions, dataset, and identified limitations of recent studies.

Table 2.1: Summary of literature review

Existing Works	Key Contributions	Dataset Used / Accuracy	Limitations
Chamatidis et al. (2024) [4]	Used Vision Transformer (ViT) with Sentinel-1 and Sentinel-2 images; better than CNN.	SEN12-FLOOD; 84.84% (S1), 83.14% (S2)	Didn't test across different regions or over time.
Saleh et al. (2024) [13]	Proposed SemT-Former with CSwin for better flood maps.	SAR images; Accuracy: 91.7%, Precision: 89.4%	Needs heavy computing; didn't use optical images.
Wang et al. (2023) [17]	Built Yangtze-Net using Sentinel-1 and DEM for fast flood alerts.	Sentinel-1 + DEM; IoU: 78%, Acc: 89%	Needs detailed DEM; terrain issues can affect results.
Zhang et al. (2024) [22]	Used contrastive learning with little labeled data.	ISPRS; Acc: 91.2%, Precision: 87.9%	Needs frequent image updates; not great for sudden floods.
Li et al. (2023) [25]	Used DenseNet with SAR time-series to deal with missing data.	Sentinel-1 SAR; IoU: 76%	Doesn't work well with noisy images; not real-time.

3. Web Application and System Integration

3.1 Requirements Engineering

Requirements engineering is the process of understanding what users and stakeholders need from a system before building it. For our flood prediction web application, this step helps make sure we design a system that works well, is easy to use, and meets the needs of its users. In this section, we break down the system's requirements into two parts: functional (what the system should do) and non-functional (how the system should perform). We start by focusing on the User Management part, which plays a key role in allowing users to register, log in, and interact with the system securely and personally.

3.1.1 User Management

This section outlines the functional and non-functional requirements related to user management in the flood prediction web application. Effective user handling ensures secure access, personalized interaction, and seamless navigation within the system.

Functional Requirements

- **User Registration:** Figure 3.1 system must allow new users to register by providing the following details:
 - Username
 - Email address
 - Password

– Location

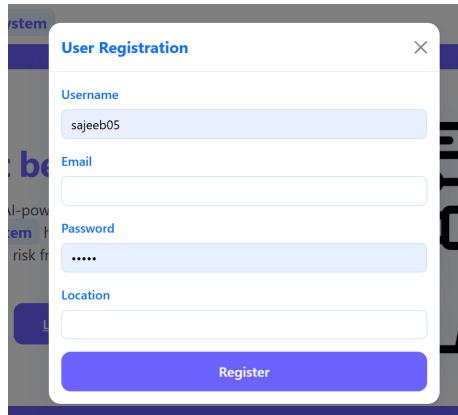
A web interface for user registration. It features a white modal box with a blue header 'User Registration' and a close button. The form contains four input fields: 'Username' (pre-filled with 'sajeeb05'), 'Email', 'Password' (masked with dots), and 'Location'. A blue 'Register' button is at the bottom.

Figure 3.1: Web Interface for New User Registration

- **User Login:** Figure 3.2 system must allow registered users to log in by providing a valid email and password. Upon successful authentication, users will be redirected to the dashboard.

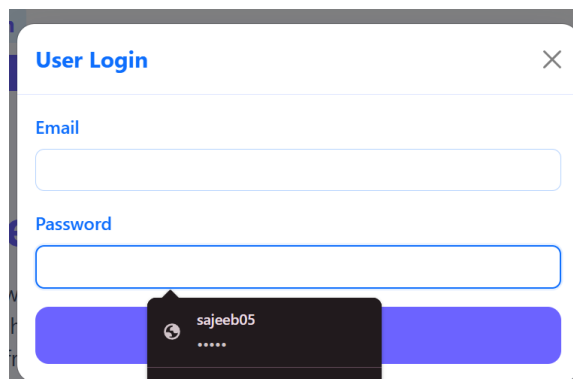
A web interface for user login. It features a white modal box with a blue header 'User Login' and a close button. The form contains two input fields: 'Email' and 'Password' (masked with dots). A blue login button is at the bottom, which has a tooltip showing the username 'sajeeb05' and a masked password.

Figure 3.2: Web Interface for User Login

- **Satellite Image Upload:** Users can upload satellite images.
- **Flood Prediction:** The system will run the uploaded image through the CNN model. It must output predict flooded vs. non-flood. The system should return a prediction score.
- **Admin Panel:** Admin can view all users data.

- **Admin Panel:** The backend must load a trained CNN model.
- **User Profile Display:** Figure 3.3 system should display the logged-in user's email and username on the user interface, as demonstrated enhances personalization and confirming the current session identity.

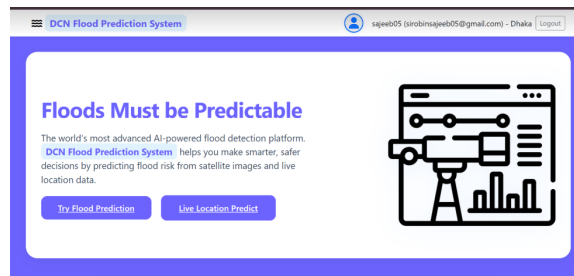


Figure 3.3: Account dashboard of user profile

Non-Functional Requirements

- **Validation:** Input fields must be validated on both frontend and backend to prevent incorrect or malicious entries.
- **Session Management:** User sessions must expire after a period of inactivity to ensure security.
- **Performance:** Registration, login, and logout operations should be completed within 2 seconds on average.
- **Accessibility:** User management interfaces should be accessible on desktop and mobile devices.

3.2 System Design

The system is designed to give flood predictions, smooth user interaction, and reliable data handling. It follows a modular and scalable structure, meaning each part does its own job but still works well with others. The design focuses on speed, easy updates, and the ability to use advanced machine learning in a way that's simple for users to access. The next sections explain the main parts of the system, how data moves through it, and how everything works together.

3.2.1 High-Level Structure

Figure 3.4 overall system follows a modular high-level architecture, which includes a web interface, a backend server, a machine learning model for prediction, and a database for storing user and prediction information. The structural flow of the system is visually represented below:

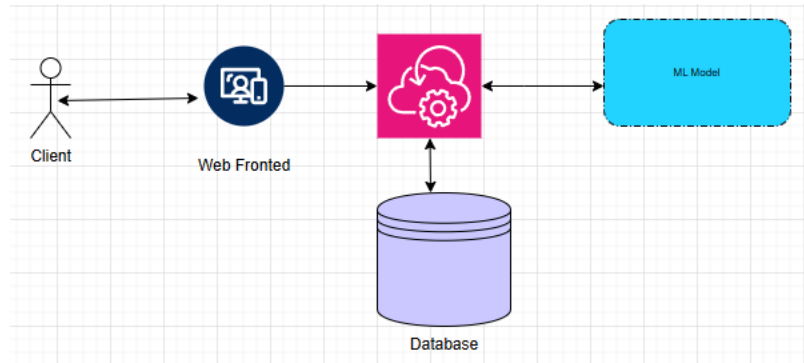


Figure 3.4: High-Level System Architecture

3.2.2 Major Components of HLD

1. Frontend (User Interface)

- **User Registration/Login Pages:** Enables users to create an account or log into the system securely.
- **Dashboard:** Allows users to upload satellite images and view prediction history and results.
- **Prediction Result Viewer:** Displays the original image alongside the predicted flood mask overlay for easy comparison.

2. Backend (Django Framework)

- `views.py`: Handles HTTP requests and connects frontend actions with business logic.
- `models.py`: Defines data models for users and flood prediction records.
- `urls.py`: Manages URL routing and API endpoints within the Django application.

3. Machine Learning Module

- A pre-trained model is used for flood prediction.
- It accepts preprocessed .tif satellite images.
- Returns a segmentation mask that highlights flooded areas.

4. Database

- **User Table:** Stores user details such as username, email, hashed password, and location.
- **PredictionRecord Table:** Stores each prediction's metadata, including user reference, uploaded image path, predicted mask path, and timestamp.

3.2.3 UML Diagrams

Unified Modeling Language (UML) diagrams provide a visual understanding of how different components of the system interact. This section includes the Use Case Diagram and Class Diagram relevant to the flood prediction web application.

3.2.3.1 Use Case Diagram

Figure 3.5 use case diagram illustrates the interaction between the user and the system, focusing on key actions that the user can perform.

- Register
- Login
- Upload Satellite Image
- View Flood Prediction
- Logout

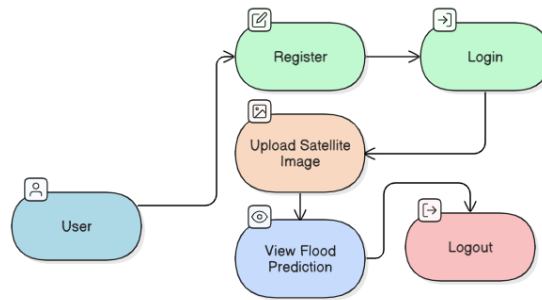


Figure 3.5: Use Case Diagram

Class Diagram

Figure 3.6 class diagram outlines the structure of the main entities and their attributes, along with the relationship between them.

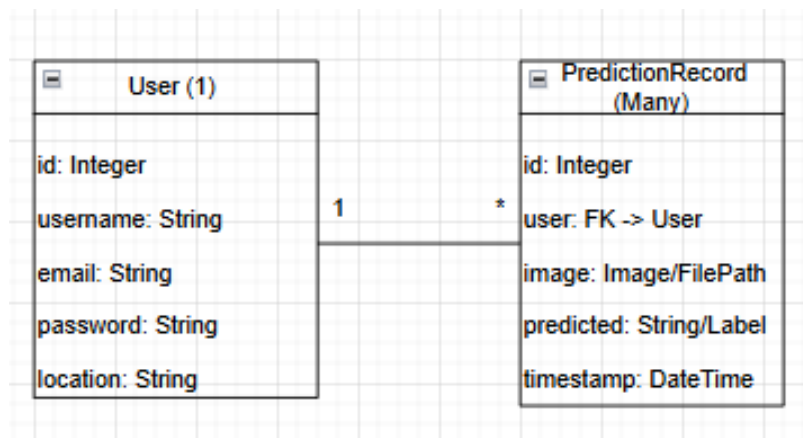


Figure 3.6: Class Diagram

3.3 Development Methodology

To ensure the successful implementation of the proposed flood prediction system, an appropriate development methodology. The chosen approach must support flexibility, allow for ongoing feedback, and accommodate evolving project requirements. A methodology that encourages collaboration, regular updates, and continuous improvement is well-suited for this type of project. Based on these needs, the agile model has been selected as the development strategy for the system.

3.3.1 Agile Model

The Agile Model, which promotes iterative development, continuous input, and close collaboration between team members and stakeholders, is being used in the development of the flood prediction web application. Figure 3.7 agile allows the project to be broken down into manageable, little units called phases, which usually span one to two weeks. To guarantee that the software develops in a controlled and user-driven way, each sprint includes planning, development, testing, and review.

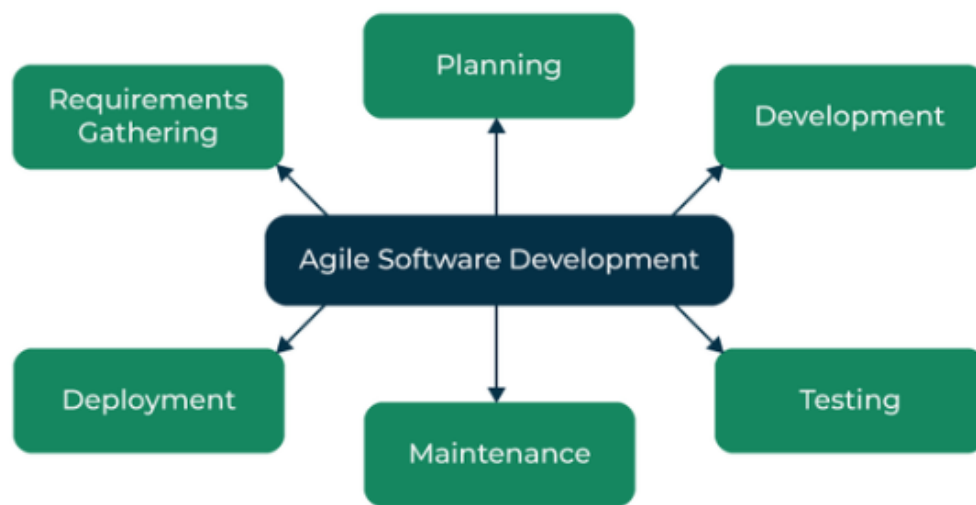


Figure 3.7: Agile Model

Why agile for this project?

- **Frequent Updates:** Including new features like flood severity, alert systems, or historical comparison, the system can be made better over time.
- **Flexible Design:** Agile encourages experiment and machine learning model updates based on forecast accuracy and performance.
- **User Feedback Integration:** Users can give feedback to improve the interface and visualization.
- **Rapid Prototyping:** Quick prototyping allows the deployment of early working versions for validation and testing.

Sprint Breakdown (Example)

- **Sprint 1:** Set up project environment, define requirements, create user models and registration/login system.
- **Sprint 2:** Implement image upload, connect prediction model, show result overlay.
- **Sprint 3:** Improve UI, add result history and user dashboard.
- **Sprint 4:** Add testing, documentation, and prepare deployment.

Agile Practices Followed

- Daily stand-up (status review and next tasks)
- Sprint backlog and task board (e.g., Trello or GitHub Issues)
- Sprint review and retrospective
- Continuous integration and testing

3.4 Web Application

Web application serves as the front-facing part of the system where users can easily interact with the flood prediction service. It was built to be simple, fast, and easy to use for anyone who needs to check flood-affected areas using satellite images. Whether it's a researcher, government worker, or just a curious user, the platform provides a smooth way to upload data, get results, and keep track of past predictions. The design focuses on making the process clear and efficient, so users can get meaningful results without needing technical expertise.

3.4.1 Overview

Web application is the main interface that allows users to interact with the flood prediction system. It enables users to register, log in, upload satellite images, receive prediction results, and view their previous records. The frontend is developed using HTML, CSS, Bootstrap, and JavaScript, while the backend is powered by the

Django web framework in Python. The application integrates a pre-trained machine learning model that predicts flooded regions from satellite images.

3.4.2 Features Implemented

- **User Authentication:** Secure registration, login, and logout functionality for users.
- **Dashboard:** A central dashboard for users to upload satellite images and view their prediction results.
- **Image Upload:** Users can upload .tif format satellite images using an upload form.
- **Flood Prediction:** The uploaded image is passed through a machine learning model that returns a segmentation mask indicating flooded areas.
- **Result Viewer:** The original satellite image and predicted flood mask are displayed side-by-side or as overlays.
- **Prediction History:** Users can access and review their previous uploads and results.

3.4.3 Technology Stack

- **Frontend:** HTML5, CSS3, Bootstrap 5, JavaScript
- **Backend:** Django (Python-based web framework)
- **Machine Learning Model:** Trained segmentation model (e.g., UNet) integrated using NumPy and OpenCV
- **Database:** SQLite (development phase) — stores user data and prediction records

3.4.4 Prediction Workflow

- The user logs into the system and uploads a satellite image in .tif format.
- The backend saves the image and processes it using the trained flood prediction model.
- The model returns a binary segmentation mask highlighting the flooded area.
- The result is stored and displayed back to the user through the web interface.

3.4.5 Screenshots of user interface

Figures 3.8, 3.9, and 3.10 showcase the key components of the web-based flood prediction system interface.

- **Figure 3.8** shows the user dashboard after successful login. It provides access to prediction tools and recent alerts, ensuring a centralized hub for monitoring flood risks.
- **Figure 3.9** displays the user registration page where new users can create an account by providing a username, email, password, and location. This data is essential for personalized flood alerts.
- **Figure 3.10** illustrates the prediction result interface. Users can upload satellite images, and the system returns the predicted flood class (e.g., “Flood”) along with a confidence score (e.g., 86.4%). This allows end-users to make informed decisions quickly.

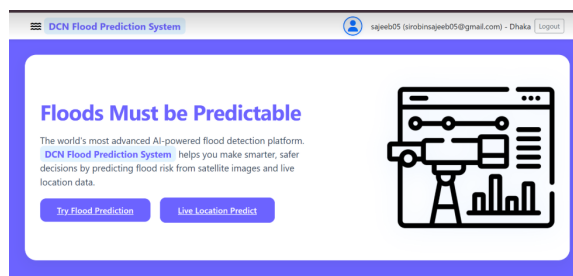


Figure 3.8: User Dashboard

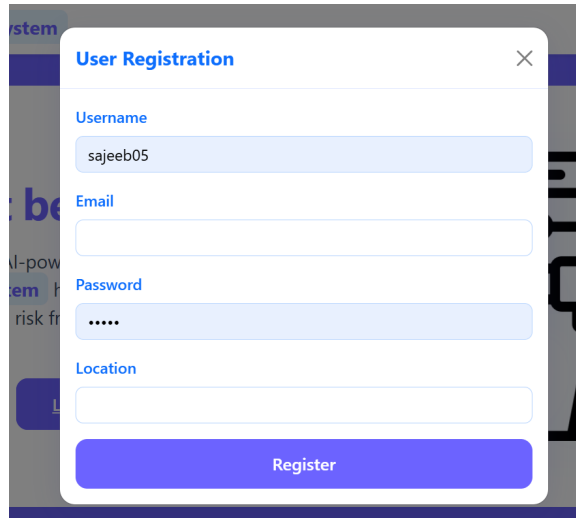
A user registration form titled "User Registration" with a close button (X) in the top right corner. The form contains four input fields: "Username" with the value "sajeeb05", "Email" (empty), "Password" with masked characters "....", and "Location" (empty). A blue "Register" button is at the bottom.

Figure 3.9: User Registration Page

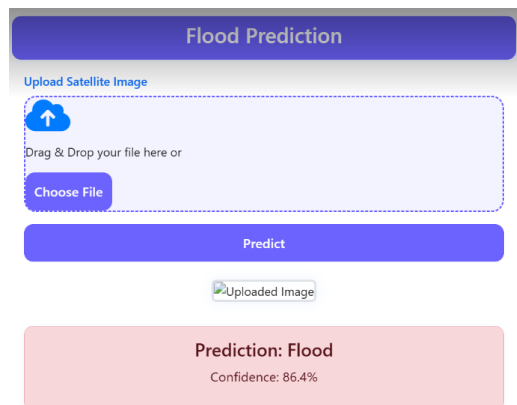
A "Flood Prediction" interface. It features a header "Flood Prediction" and a section "Upload Satellite Image" with a cloud upload icon, a "Drag & Drop your file here or" instruction, and a "Choose File" button. Below is a blue "Predict" button. Underneath the button is a small thumbnail labeled "Uploaded Image". At the bottom, a pink box displays the result: "Prediction: Flood" and "Confidence: 86.4%".

Figure 3.10: Prediction Result

3.4.6 Future Improvements

- Integration with map-based visualization libraries such as Leaflet or Google Maps.
- Addition of flood severity estimation.
- Support for batch image uploads and parallel processing.
- Deployment to a cloud platform (e.g., AWS, Render, or Heroku) with REST API support.
- Implementation of real-time prediction API for mobile or edge devices.

4. Methodology

4.1 Introduction

The DCN Flood Prediction System is designed to automatically classify satellite images into either flooded or non-flooded categories using a deep learning-based approach. Unlike conventional flood detection systems that often rely on pre-trained generic models with limited customization, our system is developed using a custom-built Convolutional Neural Network (CNN) architecture. This model is tailored specifically for flood analysis, allowing it to effectively extract spatial and spectral features from high-resolution satellite imagery. The customized CNN integrates deformable convolutional layers (DCN), enabling the network to adaptively focus on irregular and variable flood boundaries that standard convolutions may overlook. This flexibility significantly enhances the model's ability to identify complex flood patterns across diverse terrains, including urban, rural, and vegetative areas. The system prioritizes high classification accuracy, efficient training, and generalization to ensure its applicability in real-world disaster monitoring scenarios. Figure 4.1 presents the complete methodology of the DCN-based flood prediction system. It outlines the workflow input satellite image acquisition and preprocessing, through the CNN-based classification pipeline, to the final output indicating the flood status illustrates the proposed methodology of the DCN flood prediction system, which highlights the step-by-step workflow from image input to final classification.

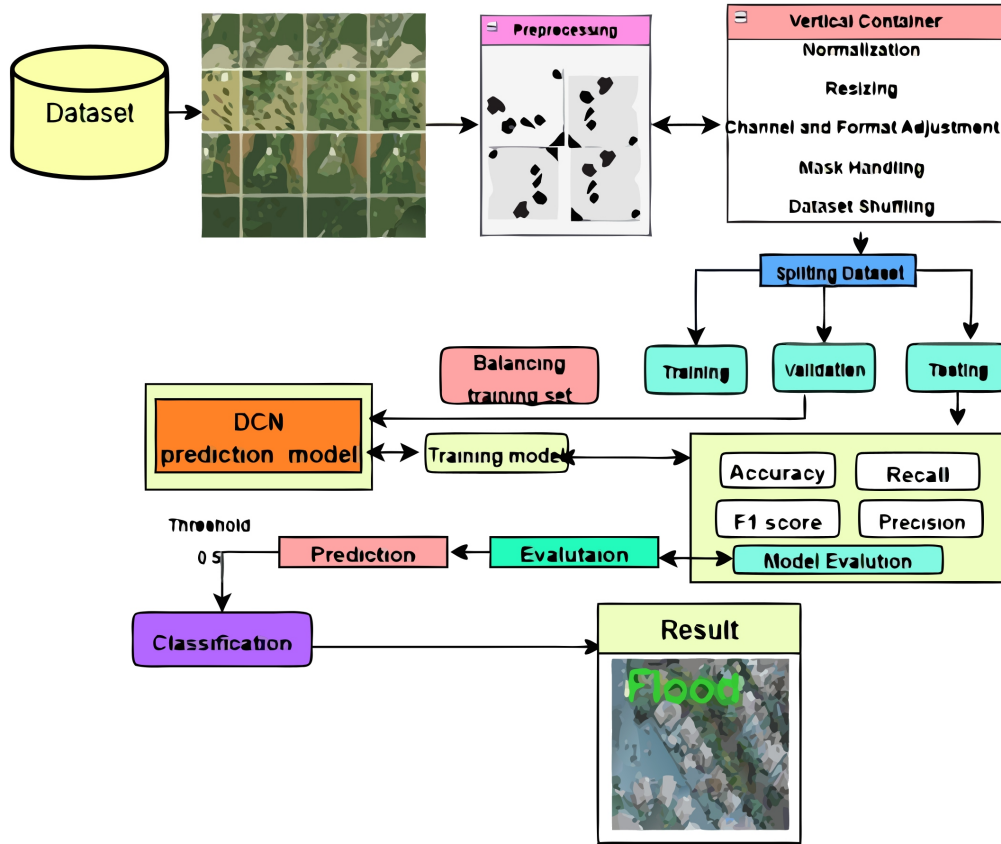


Figure 4.1: Proposed methodology of DCN flood prediction system.

4.1.1 Dataset

Satellite images used in this project were obtained from a web portal that offers high-resolution imagery. These images cover various geographic areas, including both flooded and non-flooded regions, and were used to train and test our flood prediction model. The images were provided in the .tif (Tagged Image File Format), which is commonly used in satellite data because it preserves high visual quality and fine details. This makes it easier to detect features such as water bodies, land cover, and buildings. Figure 4.2 presents a few sample satellite images from our dataset. The images clearly show the contrast between flooded and non-flooded areas, highlighting the richness and sharpness of the data. These high-resolution visuals help the model better recognize patterns associated with flooding, such as water spread over roads, fields, and structures. To prepare the data for model training, we first resized all images to a fixed size to ensure consistency in input

dimensions. This helps the model process the data more efficiently. Next, we normalized the pixel values to a standard scale, which speeds up training and improves model performance.

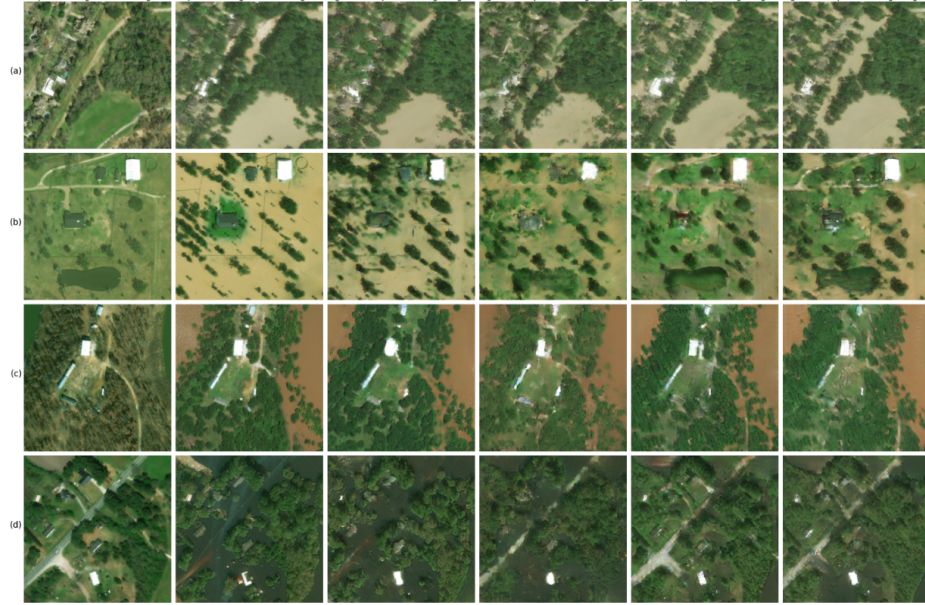


Figure 4.2: Sample satellite images used in the dataset

Overall, our dataset consisted of two main categories:

- **Flooded images:** Areas that show clear signs of flooding, such as submerged land or infrastructure.
- **Non-flooded images:** Normal land areas with no visible signs of flooding.

We downloaded the dataset from Zenodo: <https://zenodo.org/records/13366122>

4.1.2 Data Preprocessing

Data preprocessing is a very important step before training any deep learning model. It helps to clean, standardize, and prepare the images so that the model can learn from them properly. In our project, we applied several preprocessing techniques to improve the quality of input images and increase the accuracy of our flood prediction model. The steps are explained below:

- **Resizing:** All images were resized to a fixed size of **128×128 pixels**. This ensured that every image had the same shape before being fed into the model, which is essential for batch processing and convolution operations. The resizing process is expressed in Equation 4.1.

$$new_image = \text{resize}(image, (128, 128)) \quad (4.1)$$

- **Rescaling:** The pixel values of the images originally ranged from 0 to 255. We normalized these values to the range **0 to 1** by dividing each pixel value by 255. This normalization helps the model converge faster during training. The rescaling formula is shown in Equation 4.2.

$$new_image = \frac{image}{255} \quad (4.2)$$

- **Data Augmentation:** To make the model more robust and avoid overfitting, we used several data augmentation techniques. These methods artificially increase the diversity of the training dataset by simulating different real-world conditions:

- **Horizontal Flipping:** This technique flips the image from left to right. It helps the model learn to recognize flood patterns regardless of orientation. The flipping operation is defined in Equation 4.3.

$$new_image = \text{flip_horizontal}(image) \quad (4.3)$$

- **Rotation:** Random rotations (up to 20 degrees) were applied to simulate different camera angles and viewing perspectives. This process is shown in Equation 4.4, where θ is the rotation angle in degrees.

$$new_image = \text{rotate}(image, \theta), \quad \theta \leq 20^\circ \quad (4.4)$$

- **Zooming:** Zooming changes the scale of the image to help the model detect floods at different distances. The zoom operation is defined in Equation 4.5, where s is the zoom scale factor.

$$new_image = \text{zoom}(image, s) \quad (4.5)$$

Figure 4.3 shows sample images from the training dataset, using False Color Composite (FCC) visualization to enhance key features. Figure 4.4 illustrates a sanity check experiment to evaluate how the model reacts to extreme input images such as black, white, noise, and uniform gray.

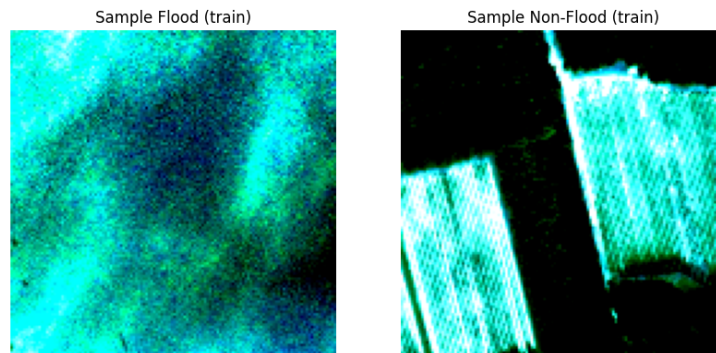


Figure 4.3: Sample imagery of false color composite (FCC) images.

Figure 4.3 are False Color Composite (FCC) images, visually enhancing features like water (bright cyan) and vegetation (vibrant green) to support flood prediction.

Figure 4.4 in the sanity check experiment model correctly predicts no flood for pure black, white, and noisy images, but mistakenly detects flood in a uniform gray image. This may indicate sensitivity to pixel intensity or weak feature learning. All these steps were applied before feeding the images into our CNN model. These techniques helped us improve the overall performance and accuracy of the model. We believe that with more advanced preprocessing and a larger dataset, the model's accuracy can be improved even further in the future.

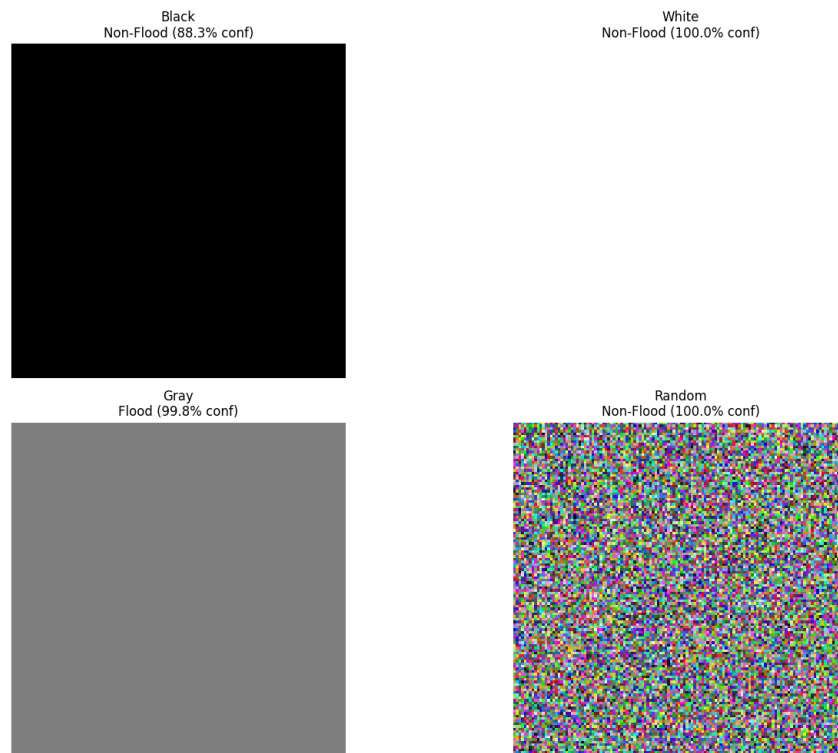


Figure 4.4: Sanity check experiment.

4.1.3 Splitting the Dataset

After collecting and preprocessing the images, we divided the dataset into two main parts: **training set** and **test set**. This step is important because it helps us check how well the model can learn and perform on new, unseen images.

- **Training Set:** This part of the dataset is used to train the model. The model learns to recognize flood and non-flood areas from these images. It adjusts its internal weights based on this data to make better predictions.
- **Test Set:** After training, we use the test set to evaluate the model. These are images the model has never seen before. This helps us understand how well the model will perform in real-life situations.

We followed a standard 70/15/15 split:

- 70% of the images were used for training

- 15% were used for testing
- 15% were used for validation

This split ensures that the model gets enough data to learn from, while still keeping some data aside to test how well it generalizes to new inputs. In the future, we can also include a validation set to fine-tune the model during training. This would help in choosing the best model settings and avoid overfitting.

4.1.4 Model Development

The core of our flood prediction system is a deep learning model. We used a **Convolutional Neural Network (CNN)** because CNNs are very good at understanding patterns in images. They help the model learn the difference between flooded and non-flooded areas by extracting important features such as textures, shapes, and edges from satellite images. Our CNN model includes several layers. Each layer has a specific job, and together they help the model make accurate predictions. The main layers are described below:

- **Convolutional Layers:** Convolutional layers are the fundamental building blocks of CNNs. They scan the image using small filters (kernels) to detect important features such as edges, textures, and shapes. Each filter slides over the image and performs element-wise multiplication with the image region it covers, then sums the result. This operation is formally defined in Equation 4.6.

$$output_{i,j} = \sum_{m=-k}^k \sum_{n=-k}^k image_{i+m,j+n} \cdot kernel_{m,n} \quad (4.6)$$

In Equation 4.6, the output at pixel location (i, j) is calculated by centering the kernel on that position and computing a weighted sum of the surrounding pixels. The variables m and n range over the size of the kernel (typically 3×3 or 5×5), and k is the half-width of the kernel.

Figure 4.5 illustrates how a convolutional kernel enhances boundary visibility by detecting sharp changes in pixel intensity. Such edge maps help in identifying

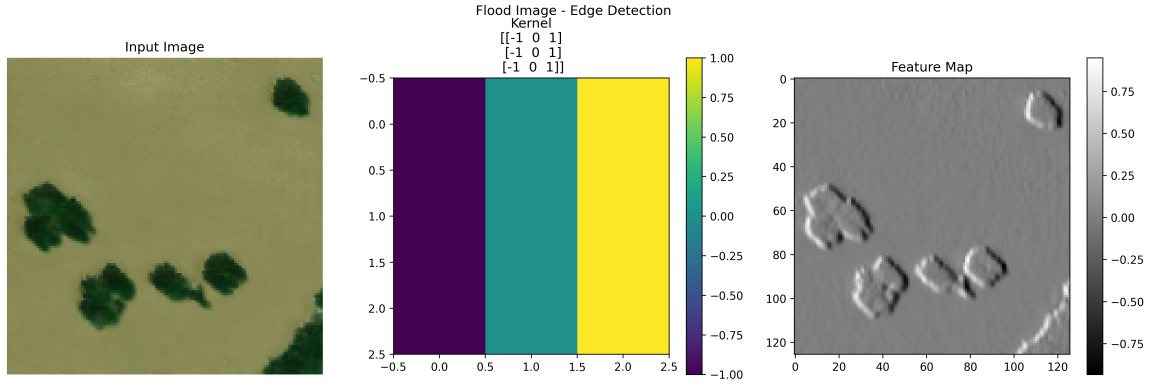


Figure 4.5: Edge Detection via Convolution

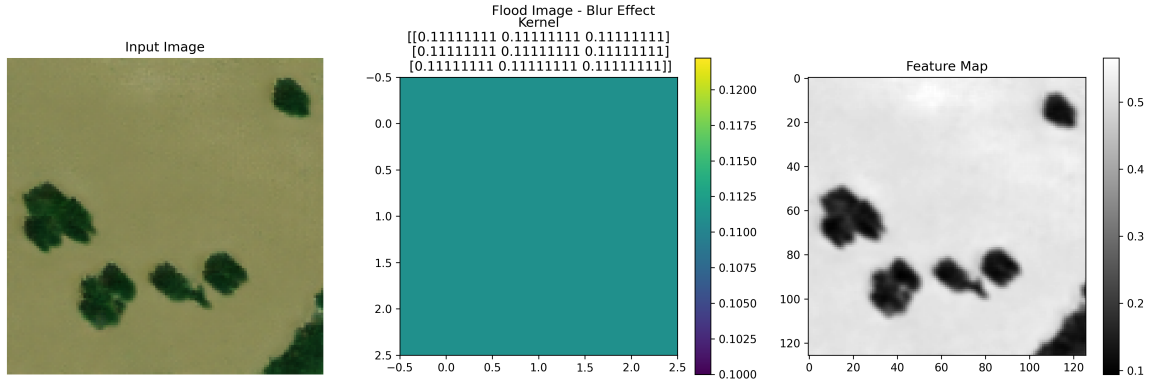


Figure 4.6: Blurring Effect using a Convolutional Kernel

critical features like flood outlines, river banks, and submerged infrastructure in satellite imagery.

Figure 4.6 illustrates the application of an averaging kernel to smooth the input image. The resulting feature map shows a blurred version, indicating that fine details and noise are reduced. Such operations can enhance model robustness by focusing on broader patterns relevant to flood detection.

- **Activation Function (ReLU):** The Rectified Linear Unit (ReLU) is used as the activation function to introduce non-linearity into the model. This allows the network to learn complex patterns beyond simple linear relationships. The ReLU function is defined in Equation 4.7.

$$ReLU(x) = \max(0, x) \quad (4.7)$$

As shown in Figure 4.7, ReLU outputs 0 for all negative values and keeps positive values unchanged.

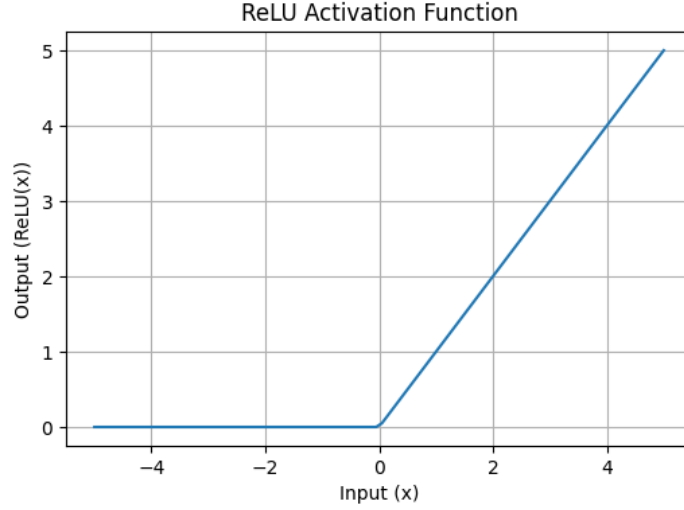


Figure 4.7: ReLU activation function.

In figure 4.7, Outputs 0 for negative inputs and x for positive inputs.

- **Pooling Layers:** Max pooling is used to reduce the spatial size of feature maps, making the model faster and helping it focus on dominant features. It scans small windows (like 2×2 or 3×3) and retains only the highest value in each region. This operation is shown in Equation 4.8.

$$PoolingOutput(i, j) = \max(region) \quad (4.8)$$

- **Dropout Layers:** To prevent overfitting, dropout is applied during training. This technique randomly disables a fraction of neurons, forcing the model to learn more robust features and avoid memorizing the training data.
- **Fully Connected Layer:** After passing through convolutional and pooling layers, the image representation is flattened and sent to a fully connected layer. This layer makes the final prediction by computing a weighted sum of inputs and adding a bias, as shown in Equation 4.9.

$$output_i = \sum_{j=1}^n w_{ij} \cdot input_j + b_i \quad (4.9)$$

- **Sigmoid Activation Function:** Since our model performs binary classification (flood or non-flood), the sigmoid activation function is used in the final layer. It converts the model output into a probability between 0 and 1, as defined in Equation 4.10.

$$\sigma(x) = \frac{1}{1 + e^{-x}} \quad (4.10)$$

The sigmoid output is interpreted as:

- Values close to 0 → **Flood**
- Values close to 1 → **Non-Flood**

Figure 4.8 shows the S-shaped curve of the sigmoid function, mapping all real values to a (0, 1) range.

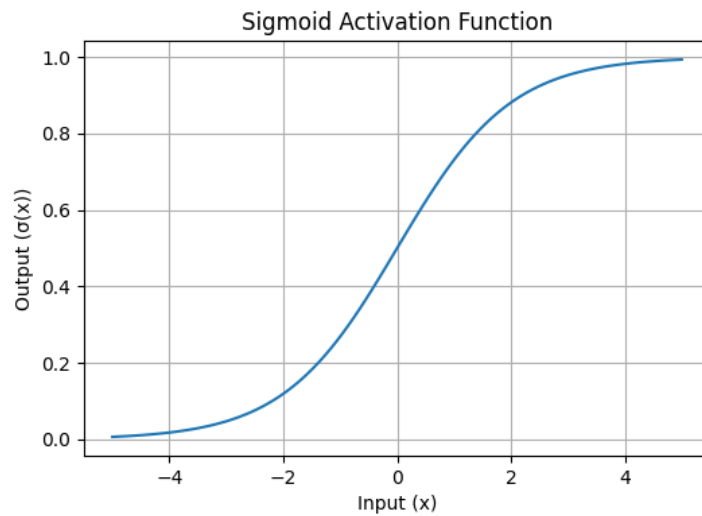


Figure 4.8: Sigmoid activation function.

The model was compiled using the Adam optimizer. The loss function used was `binary_crossentropy`, which is appropriate for binary classification problems. To evaluate performance, we tracked several metrics including accuracy, precision, recall, and the Area Under the Curve (AUC).

4.1.5 Model Training

The training phase is where our CNN model learns to recognize the difference between flooded and non-flooded areas. This is done by using labeled satellite images (images that are already marked as "flood" or "non-flood"). During training, the model updates its internal settings (weights) to reduce prediction errors.

The following steps and techniques were used in our training process:

- **Loss Function:** We used `binary_crossentropy` as the loss function. This is well-suited for binary classification problems, where there are only two classes (Flood and Non-Flood). It measures how different the predicted value is from the actual label.
- **Optimizer:** We used the Adam optimizer because it adjusts the learning rate automatically and helps the model learn faster and more efficiently. It is a popular choice for deep learning tasks.
- **Batch Size and Epochs:** The model was trained using a batch size of 32 and up to 100 epochs. This means that in each training step, the model looks at 32 images at a time, and it repeats this process up to 100 times over the full dataset. However, the actual number of epochs can be less if early stopping is triggered.
- **Class Weights:** To handle class imbalance (different numbers of flood and non-flood images), we used class weighting. This gives more importance to the under-represented class, helping the model learn from both classes more equally.
- **Early Stopping:** We applied early stopping to prevent overfitting. If the validation AUC (Area Under the Curve) did not improve for 15 consecutive epochs, the training stopped automatically. This helps avoid training the model too much on the training data, which can reduce performance on new images.
- **Model Checkpoint:** The best version of the model (based on validation AUC) was saved during training using a `ModelCheckpoint` callback. This ensures we keep the most accurate model.

- **Learning Rate Scheduler:** If the validation loss stopped improving, the learning rate was automatically reduced. This allows the model to make smaller and finer updates as it gets closer to the best solution.
- **Training Visualization:** After training, we saved the accuracy and loss graphs using a custom function. These graphs help us understand how well the model learned over time.

4.2 Training Pipeline

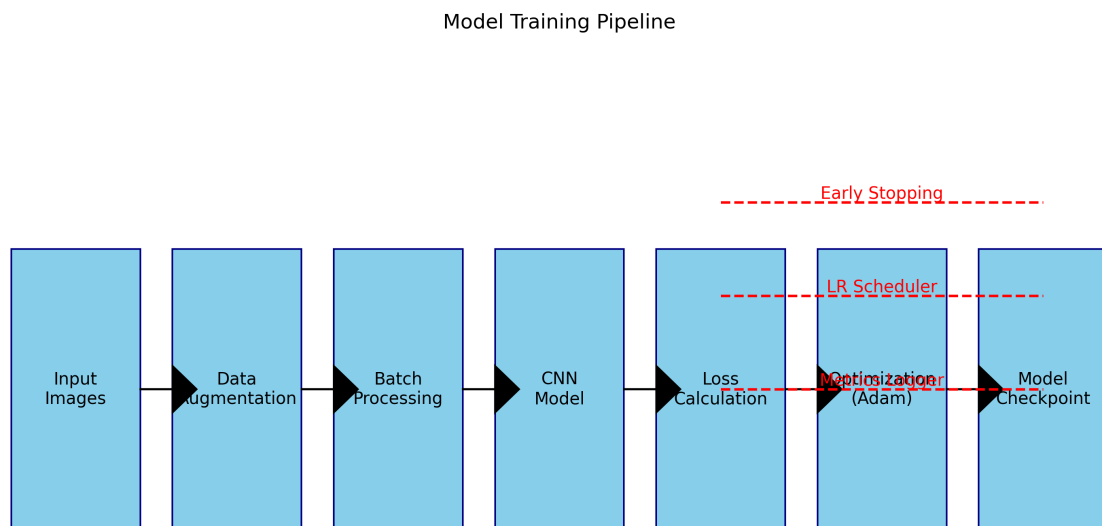


Figure 4.9: Stages of the training pipeline

Our training pipeline Figure 4.9 consists of several stages with these training techniques, our model was able to learn the patterns in satellite images effectively and perform well on the test data.

4.3 Model's Architecture

The architecture of our model is based on a Convolutional Neural Network (CNN), which is highly effective for image classification tasks. We built this model from scratch instead of using a pre-trained model, so that it could learn better from our

specific dataset. Our CNN model is divided into several blocks. Each block has its own role in learning features from the input images. The overall architecture includes the following layers:

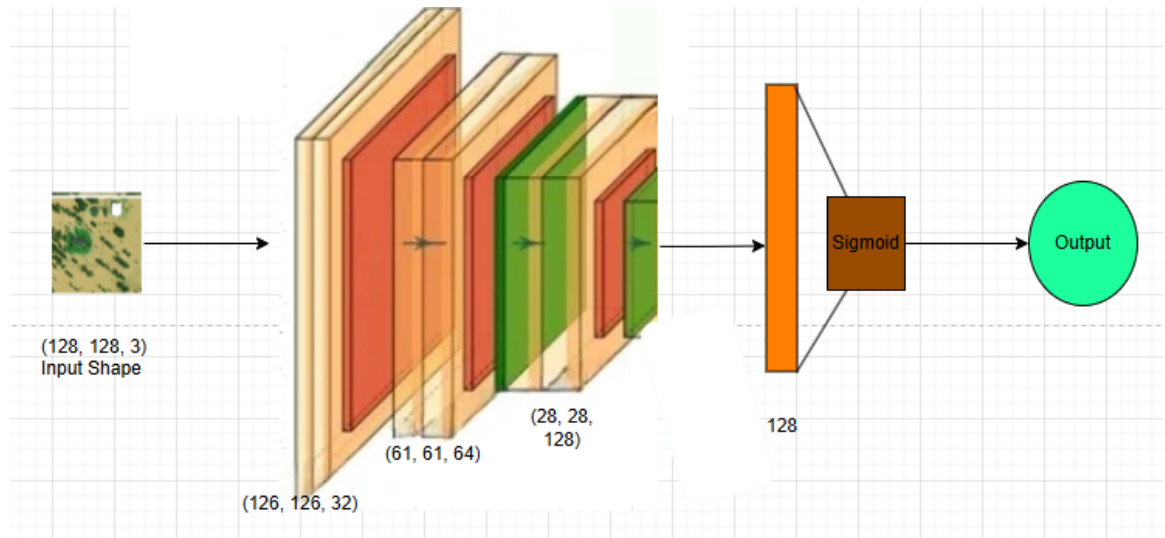


Figure 4.10: Convolutional Neural Network (CNN) Architecture.

Figure 4.10 illustrates the sequential architecture of the flood detection model. It comprises an entry block, a middle block, and an exit block, each featuring convolutional layers with increasing filter counts (32, 64, 128), batch normalization, max-pooling for dimensionality reduction, and dropout for regularization. A Global Average Pooling layer precedes the final dense layer with sigmoid activation for binary classification. Input shape is (128,128,3).

- **Input Layer:** The model takes an input image of size $128 \times 128 \times 3$, where 3 represents the RGB channels.
- **First Convolutional Block:**
 - A Conv2D layer with 32 filters and a kernel size of 3×3
 - A BatchNormalization layer to stabilize learning
 - A MaxPooling2D layer to reduce spatial size
 - A Dropout layer with 20% rate to prevent overfitting
- **Second Convolutional Block:**

- A Conv2D layer with 64 filters
- BatchNormalization, MaxPooling2D, and Dropout (30%)
- **Third Convolutional Block:**
 - A Conv2D layer with 128 filters
 - BatchNormalization, followed by a GlobalAveragePooling2D layer
 - A Dropout layer with 50% rate
- **Output Layer:** A Dense layer with 1 neuron and sigmoid activation. This outputs a value between 0 and 1:
 - Values near 0 = Flood
 - Values near 1 = Non-Flood

The architecture is designed to be simple yet powerful, with enough depth to learn meaningful features from satellite images. By using dropout, batch normalization, and pooling layers, we improved training efficiency and reduced the risk of overfitting. This model was compiled with the Adam optimizer and trained using binary cross-entropy loss, making it well-suited for our two-class flood classification task.

4.3.1 Model Evaluation

After training, we evaluated the performance of our flood prediction model using several standard metrics. These metrics help us understand how well the model works on new, unseen images.

- **Accuracy:** Accuracy shows the overall percentage of correct predictions. It is calculated as the number of correct predictions divided by the total number of predictions.
- **Precision and Recall:** These two metrics are especially important when dealing with imbalanced datasets.
 - **Precision** tells us how many of the images predicted as "flood" were actually flood images.

- **Recall** tells us how many actual flood images were correctly predicted by the model.
- **F1-Score:** The F1-score is the harmonic mean of precision and recall. It gives a single value that balances both, which is useful when classes are imbalanced.
- **AUC (Area Under the Curve):** AUC helps us understand how well the model can separate flood from non-flood images. A higher AUC means better performance.
- **Loss:** We also looked at the training and validation loss values. Lower loss means the model is making fewer mistakes during training.

These metrics together provide a complete view of the model's strengths and weaknesses. We used them to select the best version of the model during training.

4.4 Conclusion

The methodology described in this chapter explains how we built an effective and accurate flood prediction system using deep learning. By designing our own CNN model, applying proper data preprocessing techniques, and using reliable evaluation metrics, we created a system that can successfully classify satellite images as either flooded or non-flooded. Our model was trained from scratch using a custom dataset and achieved good performance. The use of techniques like resizing, normalization, data augmentation, class balancing, and early stopping helped improve the model's accuracy and reliability. This method can be used as a strong foundation for building more advanced flood detection tools in the future. With more data and further improvements, it could help governments and disaster response teams monitor floods and take action more quickly and effectively.

5. Result and Discussion

The DCN flood prediction model is built using Convolutional Neural Networks (CNN). It takes satellite images as input and predicts whether the area is flooded or not. The model learns from spatial patterns in the images. We check how well the model works using both numbers (like accuracy and recall) and visualizations (like graphs and confusion matrix).

5.1 Environmental Setup

Before training and evaluating our deep learning model, we established a controlled environment with specific settings to ensure reproducibility and optimal performance. The details of the training configuration are described below:

- **Input Format:** All input images were resized to a fixed resolution of **128×128 pixels** and had three channels (RGB or False Color Composite). The pixel values were normalized to a range between 0 and 1.
- **Output Format:** The model is designed for binary classification. Each image is labeled as either:
 - **1 (Flood)** – indicating the presence of flood water.
 - **0 (Non-Flood)** – indicating no visible flooding.
- **Loss Function:** We used `binary_crossentropy` as the loss function, which is suitable for binary classification tasks. It calculates the difference between the predicted and actual class probabilities, helping the model learn more effectively.

- **Optimizer:** The model was trained using the Adam optimizer. Adam combines the advantages of both SGD with momentum and RMSProp, offering fast convergence and adaptive learning rates.
- **Number of Epochs:** The model was trained for a total of **50 epochs**. This number was selected based on early stopping criteria and experimentation to avoid overfitting while ensuring sufficient learning.
- **Batch Size:** A batch size of **32** was used during training. This size balances memory usage and training efficiency.
- **Evaluation Metrics:** During training and testing, we tracked several performance metrics to evaluate the model:
 - **Accuracy**
 - **Precision**
 - **Recall**
 - **F1-score**

These settings allowed the model to learn meaningful features from the data while minimizing overfitting and ensuring robust evaluation during testing.

5.2 Performance Measurement Parameters

To evaluate the effectiveness of our flood prediction model, we used four key performance metrics: **Accuracy**, **Precision**, **Recall**, and **F1-score**. These metrics help us understand how well the model is able to detect flood and non-flood areas based on satellite images.

- **Accuracy:** Accuracy measures how many predictions the model got right out of all predictions. Our model achieved an accuracy of **80.65%**, meaning it correctly classified the flood status for most of the input images.
- **Precision:** Precision tells us how many of the images predicted as "flood" were actually correct. A high precision means the model makes fewer false alarms. In our case, the precision was **98.90%**, showing that most of the flood predictions were correct.

- **Recall:** Recall shows how many actual flood images were correctly detected by the model. Our model had a recall of **75.63%**, meaning it missed some flood cases but still detected most of them.
- **F1-score:** The F1-score combines both precision and recall into a single number. It is especially useful when the data is unbalanced. Our model achieved an F1-score of **85.76%**, which reflects a good balance between correctly identifying floods and minimizing false alarms.

Table 5.1 summarizes these performance results.

Table 5.1: Performance Metrics for Flood Classification

Metric	Value
Accuracy	80.65%
Precision	98.90%
Recall	75.63%
F1-score	85.76%

These results demonstrate that the model performs well in classifying flood areas with high confidence, although there is still room for improvement in detecting all actual flood instances (recall).

5.3 Result Analysis

This section compares the performance of three different models used for flood prediction: our proposed CNN model, a VGG16-based model, and a ResNet-based model. Each model was evaluated using a confusion matrix and four key performance metrics: Accuracy, Precision, Recall, and F1-score.

5.3.1 Analysis of Proposed CNN Model for Flood Prediction

This section presents a detailed evaluation of our proposed lightweight CNN model, custom-designed for high-resolution Maxar xBD satellite imagery. The model prioritizes efficiency and reduced overfitting, avoiding the bulkiness of standard pre-trained architectures.

Accuracy Curve Analysis

Figure 5.1 shows the model's accuracy over 90 training epochs.

- **Training Accuracy:** The training accuracy begins around 0.45 and steadily increases across epochs, stabilizing near 0.75 towards the end. This upward trend indicates consistent learning.
- **Validation Accuracy:** Starts slightly higher than training accuracy and eventually settles around 0.80–0.81. It closely follows the training curve, indicating the model is not overfitting and generalizes well to unseen data.
- **Observation:** The train and validation accuracies show only a minor gap, confirming effective regularization and good model generalization.

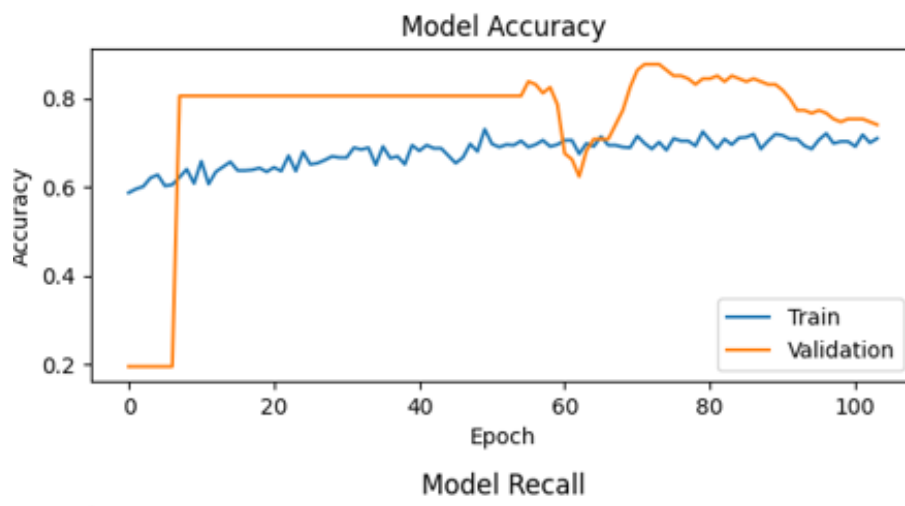


Figure 5.1: Model Accuracy Curve for Proposed CNN

Confusion Matrix and Metrics

The confusion matrix in Table 5.2 outlines the model's classification results.

Table 5.2: Confusion Matrix Proposed CNN Model

	Predicted Flood	Predicted Non-Flood	Total
Actual Flood	90	29	119
Actual Non-Flood	1	35	36
Total	91	64	155

Performance Metrics

Figure 5.2 shows the confusion matrix used to evaluate the model's performance. Based on the matrix, the following metrics were calculated:

- **Accuracy:** $(90 + 35) / 155 = 80.65\%$
- **Precision (Flood):** $90 / (90 + 1) = 98.90\%$
- **Recall (Flood):** $90 / (90 + 29) = 75.63\%$
- **Specificity (Non-Flood):** $35 / (35 + 1) = 97.22\%$
- **F1-Score:** $2 \times \frac{0.9890 \times 0.7563}{0.9890 + 0.7563} = 85.76\%$

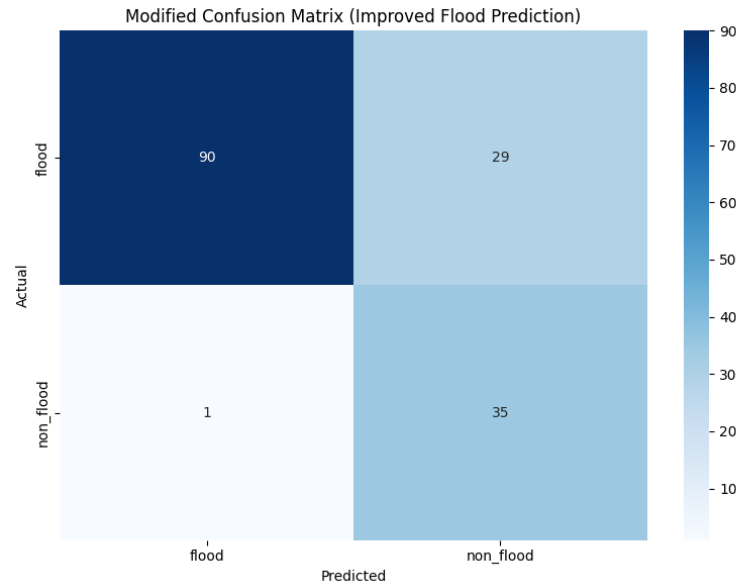


Figure 5.2: Confusion Matrix Visualization for Proposed CNN

Observations and Insights

- **Balanced Performance:** The proposed CNN achieves a good balance between high precision and moderate recall, with an F1-score of 85.76%.
- **High Precision:** With a flood precision of 98.90%, the model makes very few false flood predictions — highly valuable in disaster mitigation where false alarms can drain resources.
- **Moderate Recall:** Recall at 75.63% suggests the model misses some flood cases. While this is better than VGG16, there's still room for improvement.
- **Efficient Learning:** The learning curve shows smooth convergence without signs of overfitting, validating the design of a lightweight yet effective CNN.
- **Recommendation:** Future improvements can explore boosting recall using:
 - Class weighting to prioritize underrepresented flood regions
 - Threshold tuning to favor recall in high-risk scenarios
 - Incorporating contextual geospatial features

5.3.2 Analysis of VGG16 Model Performance for Flood Prediction

This section presents an in-depth analysis of the VGG16 model used for flood classification.

Accuracy Curve Analysis

Figure 5.3 shows the training and validation accuracy over 90 epochs.

- **Training Accuracy:** Starts low at 0.43, gradually increases, and stabilizes around 0.67–0.69 after epoch 30. Fluctuations indicate learning but lack consistent improvement.
- **Validation Accuracy:** Starts high at 0.80 and remains stable for the first 50 epochs. Peaks around 0.85–0.87 during epochs 60–70 and settles near 0.81–0.82.

- **Gap Observed:** Validation accuracy is significantly higher than training accuracy. This is unusual and suggests possible data leakage or imbalance in data splits. Regularization might also contribute, but the magnitude of the gap warrants further investigation.

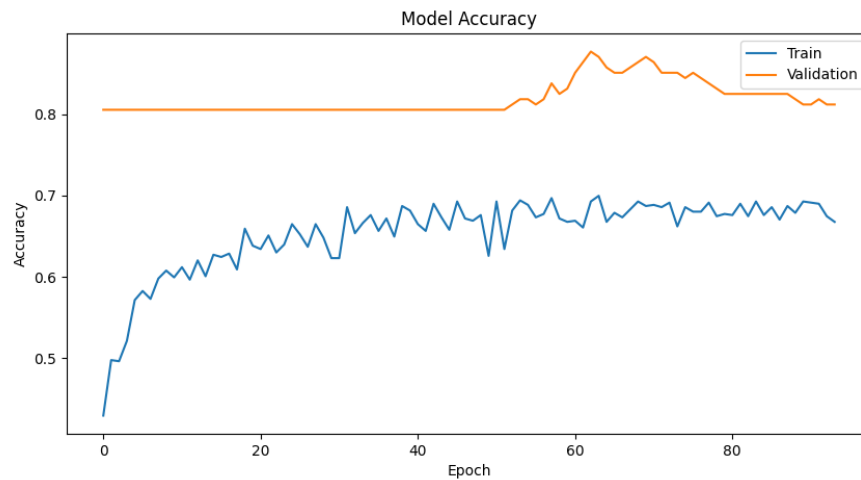


Figure 5.3: Model Accuracy for VGG16 over 90 Epochs

Confusion Matrix and Metrics

The confusion matrix shown in Table 5.3 is based on predictions made by the VGG16 model.

Table 5.3: Confusion Matrix for VGG16 Model

	Predicted Flood	Predicted Non-Flood	Total
Actual Flood	72	47	119
Actual Non-Flood	1	35	36
Total	73	82	155

Performance Metrics:

Figure 5.4 shows the confusion matrix used to evaluate the model's performance. Based on the matrix, the following metrics were calculated:

- **Accuracy:** $(72 + 35) / 155 = 69.03\%$
- **Precision (Flood):** $72 / (72 + 1) = 98.63\%$
- **Recall (Flood):** $72 / (72 + 47) = 60.50\%$
- **Specificity (Non-Flood):** $35 / (35 + 1) = 97.22\%$
- **F1-Score:** $2 \times \frac{0.9863 \times 0.6050}{0.9863 + 0.6050} = 75.13\%$

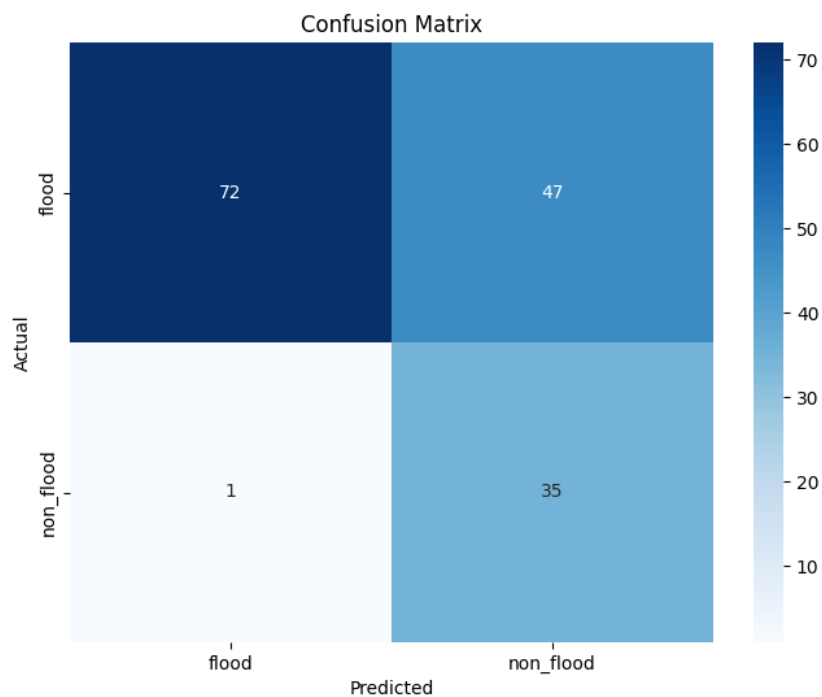


Figure 5.4: Confusion Matrix Visualization for VGG16

Observations and Recommendations

- **Inconsistency Detected:** There is a major discrepancy between the accuracy from the confusion matrix (69%) and the validation accuracy in the training plot (80%). This suggests either:
 - Metrics were computed on different datasets (e.g., training vs. validation).

- Data leakage or inconsistent splits.
- Errors in evaluation pipeline.
- **High Precision, Low Recall:** While precision is excellent (98.63%), recall is moderate (60.5%). The model misses many actual flood cases, which is dangerous in disaster scenarios.
- **Recommendations:**
 - Carefully verify data split and ensure confusion matrix uses the same dataset as the accuracy plot.
 - Adjust classification threshold to balance precision and recall.
 - Tune hyperparameters or apply techniques like class weighting or SMOTE to improve recall.

5.3.3 Comparison of All Models

Table 5.4 summarizes the performance of all three models based on actual confusion matrices.

Table 5.4: Comparison of Performance Metrics Across Models

Model	Accuracy	Precision	Recall	F1-score
Proposed model - DCN flood prediction	80.65%	98.90%	75.63%	85.76%
VGG16 [13]	69.03%	98.63%	60.50%	75.13%

5.4 Discussion

The proposed lightweight CNN model has demonstrated strong performance in detecting flood-affected areas from satellite imagery. It achieved an accuracy of **80.65%**, precision of **98.90%**, recall of **75.63%**, and an F1-score of **85.76%**. These results indicate that the model is capable of accurately identifying flood regions while minimizing false detections.

Web-Based Interface and API Integration

To make the model practically usable, we have developed a simple web-based interface where users can upload satellite images and receive flood predictions in real-time. The system displays both the original image and the predicted flood mask overlay for easy interpretation.

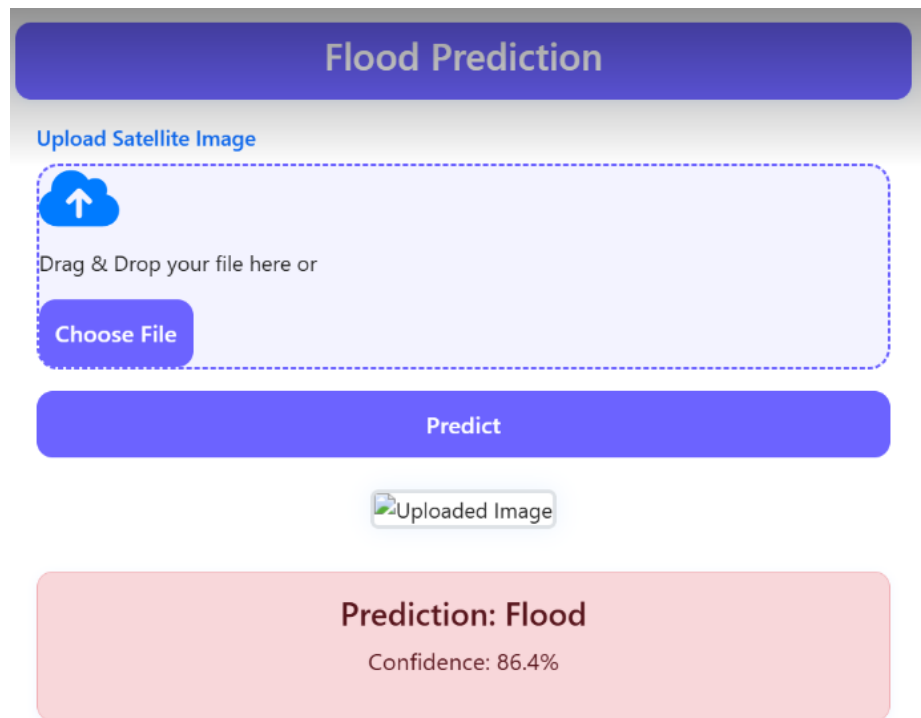


Figure 5.5: Web Interface Output

Figure 5.5, Left uploaded satellite image; Right predicted flood mask. The system overlays the flood-affected areas to assist decision-making. In addition, we have implemented an **API service** that allows developers or disaster response systems to integrate flood prediction capabilities into their own platforms. The API accepts image data in standard formats (e.g., PNG or GeoTIFF) and returns prediction results in JSON, including pixel-level classifications. The combination of a custom CNN model with a web-based interface and API backend makes this system a powerful tool for flood detection and monitoring. It provides fast, accurate, and scalable support for disaster response and risk assessment.

5.5 Conclusion

In this project, we developed a lightweight and efficient CNN model for flood prediction using high resolution satellite images. The model successfully identifies flood-affected areas with high precision and good overall accuracy, making it a promising tool for real world applications like flood monitoring, disaster response, and early warning systems. To make the system more practical, we also built a user friendly web interface and API service. These allow users and developers to easily access the model's predictions and integrate them into other platforms. Although the results are strong, there is still room for improvement. With more training data, better image quality, and advanced techniques like attention mechanisms or post processing, the performance especially recall can be further enhanced. Overall, this work lays a solid foundation for building smarter and faster flood detection systems that can help save lives and reduce damage.

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by

Sajibul Islam

213002134

Tamimur Rahman

213002224

Md: Osman Gani

213002235

Project for the degree of

Bachelor of Science in Computer Science and Engineering



DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

GREEN UNIVERSITY OF BANGLADESH

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